

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2: SYMMETRY AND RELATIVITY

TRINITY TERM 2025

Monday, 9 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Notation

Throughout the paper, c is the maximum speed for signals (the speed of light in vacuum), g is the Minkowski metric (with signature $(-1,1,1,1)$), the symbols $\mathbf{A}, \mathbf{U}, \mathbf{X}, \mathbf{P}$ indicate 4-vectors and the symbols $\mathbb{F}, \mathbb{G}, \mathbb{T}$ indicate second-rank tensors. To be precise, these symbols refer to lists of contravariant components in a column vector, such that the Lorentz transformation of a 4-vector can be written $P' = \Lambda P$ where Λ is the Lorentz transformation matrix. The dot notation for 4-vectors and tensors is defined such that

$$\mathbf{A} \cdot \mathbf{X} \equiv A^T g \mathbf{X} \equiv A^\mu X_\mu, \quad \mathbb{F} \cdot \mathbf{U} \equiv \mathbb{F} g \mathbf{U} = \mathbb{F}^{a\mu} g_{\mu\nu} U^\nu.$$

The bold font, as in $\mathbf{v}, \mathbf{E}, \mathbf{f}$, indicates 3-vectors. A pair of inertial frames S, S' with aligned axes, such that S' moves in the positive x direction relative to S at speed v , and their spacetime origins coincide, are said to be in *standard configuration*.

1. (a) Which of the following can move faster than c ? (You need not justify each answer individually, but you must provide a statement to justify your overall approach.)

- 1 A neutrino.
- 2 The wavefronts of a de Broglie wave.
- 3 The point of intersection of the blades of a pair of scissors.
- 4 A cosmic ray.
- 5 A shadow.
- 6 An interference pattern.

[3]

(b) Show that if a particle of constant rest mass m is subject to a 3-force $\mathbf{f}$, then the force, velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ are related by

$$\mathbf{f} = \lambda m \mathbf{a} + \mu m (\mathbf{v} \cdot \mathbf{a}) \mathbf{v} ,$$

where $\lambda(v)$ and $\mu(v)$ are factors which your answer should specify. Under what circumstances is the 3-acceleration in the same direction as the 3-force? (list all the cases.)

[5]

(c) A particle of charge q and mass m is launched from the origin, with initial momentum p_0 in the x direction, into a region of uniform electric field E_0 in the y direction. Show that the Lorentz factor for the particle's speed u satisfies $\gamma^2 = 1 + \alpha^2$, where α depends linearly on time t . Your answer should include an explicit formula for this α in terms of the given quantities and time.

[3]

(d) Using an energy method, or otherwise, find a formula relating γ to the height y reached by the particle. Hence, or otherwise, show that the particle reaches height y at a time given by

$$ct = \sqrt{y^2 + \eta y} ,$$

where η is a constant which your answer should specify in terms of the initial Lorentz factor γ_0 and other given quantities.

[4]

(e) Find the x -component of the particle's 3-acceleration a_x as a function of time, and give a qualitative sketch graph of $a_x(t)$. [The sketch should have the correct shape but you are not asked to find the precise location of features.]

[4]

(f) Find the fields in the inertial frame in which the particle was initially at rest, and give a qualitative sketch of the particle's spatial trajectory in this frame. [You may quote the formulae for the transformation of the fields.] Explain whether or not the trajectory is straight at large times, and whether or not it is ever vertical (i.e. aligned with the y axis) at large times.

[6]

2. (a) A monochromatic light source emits at (angular) frequency ω_0 in its rest frame. Show that in some other frame S, the frequency of the light propagating at angle θ to the direction of motion of the source is given by

$$\omega = \frac{\omega_0 \sqrt{1 - \beta^2}}{1 - \beta \cos \theta} ,$$

where β is the speed of the source relative to S. [4]

The coordinates $(c\bar{t}, \bar{x}, \bar{y}, \bar{z})$ describe an inertial frame. In this frame, there is a source propagating along the $\bar{y}$ axis in the positive direction at constant speed u , passing through a location y_0 at $\bar{t} = 0$. A detector meanwhile propagates along the $\bar{x}$ axis in the positive direction at constant speed v , passing through the origin at $\bar{t} = 0$. The source is monochromatic in its rest frame, with angular frequency ω_0 . The coordinates (ct, x, y, z) describe the rest frame of the detector, called frame S. The detector is located at the origin of S.

(b) Obtain the trajectory $(x(t), y(t), z(t))$ of the source in S, and confirm that the source passes through $(x, y, z) = (0, y_0, 0)$ at $t = 0$. [3]

(c) Hence obtain the velocity of the source in S, and show that the speed of the source relative to the detector is

$$v_{\text{rel}} = \sqrt{v^2 + u^2(1 - v^2/c^2)} .$$
 [2]

(d) Consider the light emitted at time t . Find an expression for the time t_r at which some of this light arrives at the detector, in terms of t, u, v, y_0 . Let θ be the angle, in frame S, between the light ray and the source's trajectory. Write down, in terms of t and the given quantities, a three-vector whose direction is along the trajectory of the source, and a three-vector whose direction is along the light ray from source to detector. Hence, or otherwise, obtain an expression for the quantity

$$v_{\text{rel}} c(t_r - t) \cos \theta$$

in terms of t, u, v, y_0 and $\gamma = (1 - v^2/c^2)^{-1/2}$. [6]

(e) Find the frequency ω (observed in frame S) of the light emitted towards the detector when the source crosses the x axis, in terms of ω_0, u and v . [3]

(f) The detector is an atom which has a very narrow transition which can be excited by the radiation. The rest mass of the atom is m in its ground state and $m + \hbar\omega_0$ in its excited state. Find the value of u for which the light considered in part (v) shall excite the atom. [7]

3. (a) Let t, x, y, z be coordinates in some inertial frame, and let E , $\mathbf{p}$ and $\mathbf{v}$ be the energy, momentum and velocity of a particle of rest-mass m . State which of the following quantities are 4-vectors (they are numbered so that you can identify them easily in your answer):

$$\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \\ \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix}, & \begin{bmatrix} c \\ v_x \\ v_y \\ v_z \end{bmatrix}, & \begin{bmatrix} E/c \\ p_x \\ p_y \\ p_z \end{bmatrix}, & \begin{bmatrix} mct \\ mx \\ my \\ mz \end{bmatrix}, & \begin{bmatrix} mc \\ mv_x \\ mv_y \\ mv_z \end{bmatrix}, & \begin{bmatrix} Ec \\ Ev_x \\ Ev_y \\ Ev_z \end{bmatrix}. \end{array}$$

[3]

(b) A particle of 4-velocity $\mathbf{U}$ and rest mass m is subject to a 4-force $\mathbf{F}$. Prove that if $\mathbf{F}$ satisfies $\mathbf{F} \cdot \mathbf{U} = 0$ then m is constant. A certain force is related to 4-velocity by $\mathbf{F} = q\mathbb{G} \cdot \mathbf{U}$ where q is a Lorentz-invariant constant and $\mathbb{G}$ is a second-rank tensor. Show that a sufficient and necessary condition for $\mathbf{F}$ to be rest-mass-preserving is that $\mathbb{G}$ is antisymmetric.

[5]

(c) In an inertial frame S , the electric and magnetic fields are parallel and have magnitudes E_0 , $B_0 = E_0/c$. Show that in all other frames the fields are greater than, or equal to, E_0 and B_0 respectively. In a frame S' the magnetic field is of size $B' = \sqrt{2}B_0$. Find the angle between the electric and magnetic fields in S' .

[5]

(d) The electromagnetic field tensor $\mathbb{F}$ has components

$$\mathbb{F} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}.$$

The stress-energy tensor of the field is given by

$$\mathbb{T}^{ab} = \epsilon_0 c^2 \left(-g^{a\lambda} \mathbb{F}_{\lambda\mu} \mathbb{F}^{\mu b} - \frac{1}{4} g^{ab} \mathbb{F}^{\mu\nu} \mathbb{F}_{\mu\nu} \right).$$

Find the components of $\mathbb{T}$ for the electromagnetic field described above in the frame S , where the fields are in the y direction. Interpret the components of $\mathbb{T}$ physically. Include in your answer a statement of the relationship between momentum flux and energy density.

[6]

(e) Find the stress-energy tensor in a frame S' in standard configuration with S .

[2]

(f) At low velocities, one might have expected the energy flux in S' to be the frame velocity times the energy density (as would be the case for ordinary matter, to good approximation), but this is not the case here. For $v \ll c$, explain how the energy flux in frame S' can be accounted for as $\rho \mathbf{v}$, where ρ is energy density, plus another contribution which you should describe.

[4]

4. (a) Two particles collide and emerge with unchanged rest masses. Let the incoming 4-momenta be P and Q and the outgoing 4-momenta be P' and Q' . P and P' have rest mass m_1 ; Q and Q' have rest mass m_2 . Prove that $P \cdot Q = P' \cdot Q'$ and $P \cdot Q' = P' \cdot Q$.

[3]

(b) Introduce the definitions:

$$s = -(P + Q)^2, \quad t = -(P - P')^2, \quad u = -(P - Q')^2,$$

and take $c = 1$. Find an expression for $s + t + u$ in terms of the rest masses alone.

[2]

(c) Let $\mathbf{p}$, $\mathbf{q}$, $\mathbf{p}'$, $\mathbf{q}'$ be the 3-momenta associated with the various 4-vectors in the zero momentum frame. Let the particle energies in this frame be E_1, E_2 before the collision and E'_1, E'_2 after. Show that these energies are preserved, i.e. $E'_1 = E_1$ and $E'_2 = E_2$.

[5]

(d) Let the scattering angle θ be the angle between $\mathbf{q}$ and $\mathbf{q}'$. Obtain an expression for t in terms of $\cos \theta$ and p , and show that

$$s = m_1^2 + m_2^2 + 2(E_1 E_2 + p^2).$$

Hence obtain a formula for $\cos \theta$ in terms of s, t, m_1, m_2 alone.

[6]

(e) Explain how we can use the above results to treat a collision of the type

$$p + \bar{p} \rightarrow \pi + \bar{\pi},$$

where the notation indicates an incoming proton and antiproton, with outgoing pion and anti-pion. In practice, a $p\bar{p}$ collision can result in either outgoing $\pi\bar{\pi}$ or outgoing $p\bar{p}$, with probabilities predicted by quantum field theory. Give an argument to show that for given values of the 4-momenta, the probability of obtaining each of these outcomes is Lorentz invariant.

[3]

The remainder of the question concerns free motion of a single particle.

(f) Find the worldline which maximises the proper time for a particle moving between a given pair of time-like separated events.

[2]

(g) An action S is defined by the path integral

$$S = \int_{t_1}^{t_2} L dt,$$

where t is coordinate time in some inertial frame and L is a Lagrangian function. Find a Lagrangian such that the path of stationary action, for a given pair of events, is equal to the worldline obtained in part (vi), and find the associated Hamiltonian. Write the Hamiltonian in terms of its proper variables.

[4]